

Q: So if I have a private cloud running, what changes will I have to make in the existing hardware to adopt 12c?

You can do that even if it is not an Oracle server. This database works on every operating system. 12c does not impose any restrictions on the hardware that one has to use. It is not disruptive software. It is a database designed for multi-tenancy support but it continues to have all the characteristics that were available elsewhere, and it does not impose any penalty on the hardware it runs on. On the contrary, if you are looking to consolidate different database instances into a single database, one required more hardware and resources to run them differently in the good old days. Now these can be consolidated into one. And hence, one is able to achieve the consolidation even at the hardware level.

Q: How do you pitch this database against Cybase and HANA?

There is no comparison, I think. As far as Cybase is concerned, even to beat the player just a level ahead of them in terms of ranking, they have to grow at a CAGR of 300 per cent. So there is no comparison between Oracle and Cybase in terms of market share. So is the case when it comes to its ability to run applications. In fact, it took SAP 20-21 months to port a part of its applications on to Cybase. So porting something on Cybase is quite a task. And it will be much more difficult to do this on HANA.

Now, coming to the technological differences between Cybase and Oracle, there is a reason why Oracle became such a big and important database. According to me, the main reason why Oracle overtook everyone else is because of its scalability. A major challenge that an IT manager faces is that when more and more people come on to the database, it starts getting blocked. That is a challenge that Cybase faces. Oracle decided to architect the database differently from Oracle 7 onwards. We decided to make a monolithic database where you do not escalate the locks, and because of that we were able to scale any OLTP (Online Transaction Processing) transaction. On the mainframe level, UDB does it, which is why it is a pioneer in that space.

The challenge that we faced was that we were a great monolithic database but we could not have multiple databases running on the same server. That is why Oracle decided to come out with something specifically for the cloud. We did not want to lose our true character of being an OLTP database. We solved the issue by making something we call a 'container database' and a pluggable database. So we are very different from Cybase. I think we have been able to make a mark in the database business because we marry the greatness of the monolithic database, which allows any amount of OLTP transactions to come in, with the ability to have multiple databases in the same server with 12c.

HANA is different altogether and Oracle has a lot of products to offer against it. I think HANA should have come in 6-7 years ago when the cost of memory was very high. With the cost of memory coming down, in-memory databases are losing their importance. With in-memory databases, the challenge is of integrity, as they do not adhere to all the acid properties of the databases. The challenge with such databases is that with memory prices having been slashed, Oracle's entire database can sit in the memory. So HANA was good for how things were 6-7 years ago but, today, other technologies have taken over.

Q: New technologies like Big Data and the cloud are being adopted by companies on a large scale. What is the impact that these technologies are having on the storage sector and what are the challenges with regard to cloud-based storage?

Technologies like Big Data, which essentially relate to the huge amount of data that is generated through social media and other such sources, pose a very different kind of a challenge to a company. For instance, when a company wants to make some sense out of this kind of data, it is not sure whether to look at that data as a whole, or extract the intelligence from it, and then work on it. So Oracle's approach towards Big Data is slightly different. We think that there are two kinds of data—structured and unstructured. This data that we are talking about is actually the unstructured data. So it requires cleaning and filtering, which can be done by adequate software. After the data is filtered, you give it the same treatment that you give to any other enterprise data. We are looking at simplicity in Big Data.

Storage and security of data are the biggest challenges for anyone wanting to move to the cloud. If you look at applications, there have been multi-tenanted applications from the past 10 years or so, but now people want multi-tenancy at the database level. Let's look at an example of companies A, B, and C and the HR applications deployed in those companies. They can have multi-tenancy in terms of HR applications amongst themselves, which means all three can have their separate HR systems, but previously, at the database level, all the data was sitting together in the same storage. Because of that, privacy and security was always a challenge. That is where we are trying to make a difference with 12c. People want multi-tenancy at the database level but that previously meant the increased usage of resources. We have tried to address that issue. We have kept the resource usage at the same level as an application would have in multi-tenancy format, but given the database the same privacy as a separate database is capable of offering. So we have brought in multi-tenancy at the database level. Now, one can have the data of all three companies stored in a separate storage system and, at the same time, have this system's resources talk to each one of them. 